Chapter 1

Introduction

1.1 Research context

This paper defines the research context before selecting the most appropriate methodology and strategy for the literature review which follows.

Empirical observations, supported by academic and grey literature, (Finkelhor et al., 2024; National Police Chiefs' Council, 2025) identify that reported instances of Child Sexual Abuse (CSA) are increasing, with resultant demand on policing outstripping proactive preventative capabilities.

The gap between demand and capability in both CSA and other operational areas of policing is acknowledged across the sector, generating political and organisational appetite for technological solutions, (National Police Chiefs' Council, 2026). Despite this appetite, evidence from both professional and academic sources identifes that policing is structurally complex and requires reform to better adopt technology, (Cooper, 2024), with cultural barriers further diminishing the successful use of technological solutions, (Thompson and Manning, 2021; Kassem and Erken, 2025).

These challenges are significant, yet they do not negate the potential of emerging technologies to address the identified capability gap. Advances in Natural Language Processing (NLP) technologies present an opportunity to deliver preventative safeguarding in the CSA context, acknowledging that the gap between technical possibility and operational deployment is not primarily technical but organisational.

This paper explores how to bridge this gap, drawing on four literature tracks to develop a conceptual framework capable of addressing the proposed research challenge: *How can natural language processing methods be developed and operationalised to identify precursors of contact sex offending against children, to enhance proactive safeguarding opportunities within the existing operational constraints of policing?*

This challenge consists of four thematic research areas:

1. What linguistic precursors of contact sex offending exist in offender-victim conversations?
2. Which NLP methods can be most effectively realised within policing infrastructure?
3. What implementation barriers exist within policing that may hinder NLP adoption?

4. What strategies can be developed to overcome those barriers?

1.2 Purpose and scope of the paper

This research is inherently interdisciplinary, positioned at the intersection of technical and organisational domains. Accordingly, a literature review will require a multi-track approach, with each track addressing a different aspect of the research challenge. The four tracks are as follows:

1. The first track explores the landscape of CSA and the linguistic characteristics which precede contact offending, establishing the empirical foundation for a technical solution.
2. The second track reviews the technical literature on NLP applications in safeguarding and criminal justice contexts, focusing on grooming language detection and the computational methods available for identifying linguistic precursors of contact offending.
3. The third track examines the organisational dynamics of technology adoption in policing, drawing on both academic literature and empirical evidence of structural and cultural barriers to technical innovation.
4. The fourth track addresses the ethical and governance boundaries within which any NLP solution must operate, encompassing algorithmic bias, data protection, proportionality, and the ethical context surrounding preventative safeguarding.

Each track is explored in depth, and their insights are then synthesised to identify where technical possibilities meet organisational realities. This structure enables a comprehensive understanding of both the capabilities and constraints shaping both the research problem and the practice improvement opportunity and grounding any solution in organisational and ethical realities

Chapter 2

Literature Search Strategy and Methodology

2.1 Review Methodology and Rationale

This literature review adopts a systematic integrative approach. Snyder (2019) documents a number of review types, noting that an integrative review seeks to combine multiple perspectives, synthesise those views, and to develop a new framework or perspective.

The interdisciplinary nature of this research, noted in section 1.2, necessitates a review methodology that can accommodate multiple perspectives and evidence types. An integrative design as defined by Torraco (2016) and Torraco (2005) allows for a review and synthesis across domains, building a conceptual framework which seeks to unify the disparate domains into a framework for operationalising and implementing an NLP solution into a safeguarding CSA context.

A systematic approach is layered into this integrative approach to allow for a targeted and transparent search and selection process in reviewing the evidence, ensuring that the review is both comprehensive and replicable. This systematic layer is particularly important given the breadth of this review, which spans technical, organisational and ethical domains, each with their own bodies of literature and evidence types. Whittemore and Knafl (2005) identifies that by adding a systematic approach to an integrative review, findings and subsequent synthesis across domains can lead to the delivery of evidence based practice improvements.

From an academic perspective, a doctoral literature review needs to be defensible and reproducible. Whilst the integrative approach allows for synthesis and framework development, a layered systematic approach ensures reproducibility whilst adding a counter balance to my own personal biases and my tendency to approach problems with a solutionist mindset.

2.2 Tools and software

The multifaceted nature of this review demands a robust literature and note management strategy supported by appropriate tools. Basu (2020) describes the Zettelkasten (ZK) method which can be used to manage literature sources and create links between them, supporting the synthesis process. This is furthered by Reck (2023) who describes the ZK method as a process of creating notes on individual literature sources, applying tags to those notes and then using the tagging system to create links between the notes. Both Basu (2020) and Reck (2023) describe the ZK method as the process of building a Personal Knowledge Management System (PKMS). Cheong and Tsui (2011) describes a PKMS as a structure which allows for the development of "*personal information management, personal knowledge internalization, personal wisdom creation, and interpersonal knowledge transferring*"

Reck (2023) considers the use of various software tools to support the ZK method. Following a period of experimentation with various applications, a software stack to support the ZK method for this review was selected consisting of Zotero for literature management and bibliography generation, Obsidian for note taking and synthesis and LaTeX for writing and formatting the review.

Zotero was selected for its robust literature management capabilities, including the ability to capture metadata, PDFs and notes on literature sources, and to auto generate and update a master bibliography file. Obsidian was selected for its flexibility in note taking and synthesis, allowing for the creation of a network of linked notes which can be easily navigated and searched. LaTeX was selected for its professional formatting capabilities, particularly for handling citations, figures and tables, and for its ability to draw directly from Zotero's auto-generated bibliography, thus ensuring consistency and ease of referencing.

All content is then automatically synced to a cloud store and defined content saved a second time to a GitHub repository. My use of the ZK method and the associated software stack is illustrated in figure 2.1.

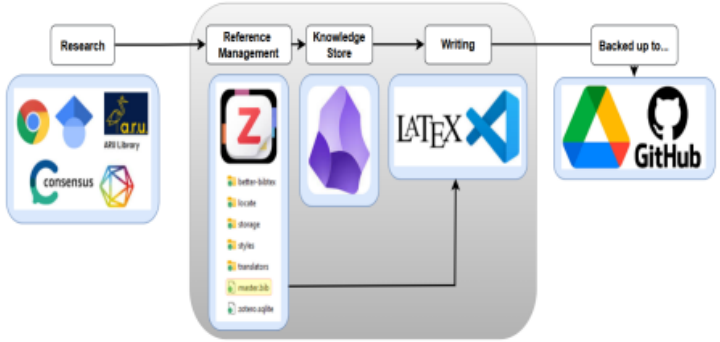

Figure 2.1: Zettelkasten method application for literature and note management

2.3 Search approach and databases

Section 1.2 identifies four distinct research tracks. Each of these tracks has a unique focus and therefore requires a dedicated search strategy to maximise the relevance and quality of included literature. The search strategy presented in table ?? defines the four core research tracks, the question to be answered for each track, the selected databases for each track, and the rationale for selecting those databases. The search strategy is designed to ensure that the literature review is comprehensive, targeted and relevant to the research challenge and that it can be replicated by other researchers. The search strategy is also designed to ensure that the literature review is manageable, given the breadth of the research challenge and the interdisciplinary nature of the review.

2.4 Keywords and search terms

Key words and search terms are critical to the success of a literature review. Chigbu et al. (2023) provides holistic guidance on keyword selection and submission to selected databases. The keywords and search terms utilised for each of the noted research tracks are defined in table ??. These align to the search strategy outlined in table ??. The keywords and search terms are designed to capture the key concepts and themes relevant to each track, ensuring that the literature retrieved is relevant to the research challenge.

Patrick Thompson

18 Mar 20:18 | Last reply 3 Apr 23:42

Hi squad!

I've uploaded a PDF entitled "Draft so far..." which is an effort to get some structure around my literature review.

I've not made much headway on actually reading lots yet as I wanted to nail down the strategy.

Hopefully the document makes logical sense. Essentially I've broken down my literature review into four tracks / topics and then decided on a systematic integrative review methodology given the four slightly separate but linked topic areas.

In the files section (https://canvas.anglia.ac.uk/groups/44479/files) I've also uploaded four tables;

One is called search_matrix. Its a table were I try to justify search sources for the four tracks.

One is called keywords. It's my attempt to narrow down keywords across those four tracks.

The other two are inclusion / exclusion criteria. The 2018 date is linked to the year when large language models became more openly available and made a big difference in the world of natural lanuage processsing.

Speak soon!

Pat

Name	Created	Last Modified	Modified by	Size	Actions
exclusion_criteria.html	18 Mar 2026	18 Mar 2026	Patrick Thompson	2 KB	
inclusion_criteria.html	18 Mar 2026	18 Mar 2026	Patrick Thompson	2 KB	
keywords_v2.html	25 Mar 2026	25 Mar 2026	Patrick Thompson	17 KB	
search_matrix.html	18 Mar 2026	18 Mar 2026	Patrick Thompson	8 KB	

Beth Brown

29 Mar 14:57

Hi Pat,

For an early first draft I thought this was awesome! Your writing is clear and you articulate the research context with precision, I felt that your opening section clearly situates the problem within policing, policy, and technological landscapes, and you explain this with a balanced, evidence-based tone.

The four research tracks are logical and well reasoned, clearly connecting to your overarching question. Your tables are also incredibly clear and organised. I think the fact you have got such a clear structure will make your literature review manageable and robust. It also shows that you've thought deeply about the complexity of the problem space.

The justification for separate strategies is sound and aligns well with the assignment brief. You could consider briefly mentioning in the introduction what is out of scope and why. This may help manage expectations and demonstrate deliberate boundary setting. Also, you've clearly explained the tracks individually but a short statement about how they will be brought together (e.g., thematic synthesis, conceptual mapping, cross-domain triangulation) would provide further clarity about the approach.

I really liked your explanation of why an integrative review is appropriate, and how you're layering systematic elements to ensure rigour. This demonstrates a strong understanding of review methodologies and defended your choices well.

Overall I thought this was great - you show really clear thinking and a strong grasp of both the technical and organisational sides of the topic! Look forward to the next Patch..

Reply | Like | Mark as unread

Lauren Hunt

3 Apr 23:42

Hi Pat,

Its reassuring we're at a similar stage, but its clear you know your topic and that you've spent a lot of time thinking of the best way to tackle it!

Love the tables in the files and numbering your in/exclusion criteria to refer to. You (and Beth) have showed your Boolean search words very well and something I'd develop in my work - I also like how you clearly showed each element and then showed the cross of the topics (even if I don't understand some of the AI bits!). Its good to see how all of your tracks meet up and relate to your question - you've done well to get into the detail of the topic but also explain each section clearly.

The search matrix took me a second to get my head around, but this only helped me as again you and Beth have used many more databases/journals than me so maybe I need to explore this more to maximise results. I was worried about over-complicating my search but you've both presented this in clear formats so I may give it another go! I can also see how your feedback to me to clear in your own work regarding in/exclusion of paper dates, with 2018 being an important time for your topic.

You justify your choices well with citations and your writing flows nicely - overall I think you've made a fantastic start and time well spent setting up a great foundation - I look forward to draft 2!

Reply | Like | Mark as unread

Patrick Thompson

24 Mar 16:11

Hi Beth,

Here are my thoughts!

The keyword choices are well considered and the reflective bit around spotting "constabulary" in the results shows non-linear thinking... you're engaging with the process rather than just running through it mechanically. The Boolean searches you settled on clearly gave you a really solid foundation of papers to work from.

My dodgy maths puts you somewhere around 5,700 returns once you focus down to policing — though with three databases there's inevitably going to be a hefty overlap, so I'm guessing the real number is probably closer to 2K.... is there scope to be more robust on the exclusion criteria? You've only got one at the moment and I wonder if tightening this would actually give you more freedom, letting you really focus on contemporary practice and lived officer experience rather than having to wade through how things have evolved over the last decade and a half?

I guess along side that is the 2010 date. Sixteen years is a long stretch and I wonder if it's worth either shortening the time span or giving that choice a really solid justification, especially since the five-year Google Scholar search felt a little inconsistent alongside it.

I really liked the thematic buckets in the screening section — they're well chosen and make sense. It did make me wonder though how you're moving from the initial screening to identifying *key* papers specifically? It might be worth a sentence on what's elevating one paper above another?

The critical summaries are cool!

Reply | Like | Mark as unread

Patrick Thompson

24 Mar 16:38

Hi Lauren,

I really like the way you've set this out — the PICO structure to define the requirements is really logical, and the dual-track in-vivo and in-vitro approach makes the methodology robust.

The inclusion and exclusion criteria are sensible and clearly reasoned. One question worth considering: does there need to be any date parameter in the search, or at least an explicit justification for why all dates are included? Reviewers may ask about this.

The planned PRISMA flow chart is a great addition and will keep the whole process transparent and reproducible. Looking forward to seeing the work develop — you seem to be at a similar stage to me in having nailed your methodology and are about to start the actual capture of content.

Reply | 1 Like | Mark as unread